CASE STUDY

Chorismate mutase (1): reaction coordinate scan

> Updated for EasyHybrid 3.2.0 (2026-10-26). The figures and numbers come from a run with the 3.2.0 development version.

Chorismate mutase (CM) catalyzes the Claisen rearrangement of chorismate to prephenate, a step of the biosynthesis of phenylalanine and tyrosine in bacteria, fungi and plants. The reaction is a concerted [3,3]-sigmatropic shift with no covalent intermediate: the C5–O7 bond of the enol ether breaks while the C1–C9 bond forms, through a chair-like transition state. The enzyme accelerates it about a million times over the reaction in water, mostly by holding the substrate in the reactive conformation and by electrostatic stabilization of the transition state.

This case study treats the same reaction three ways:

1. **a relaxed scan** of a reaction coordinate (this page);
2. a nudged elastic band (NEB) path between reactant and product, with no coordinate;
3. umbrella sampling along the coordinate, for the free energy.

We thank Dr. Atsu Agbaglo and Prof. Nathan DeYonker (University of Memphis) for the original files of this system.

The system

`examples/tutorials/case_studies/2-4_chorismate_mutase/chorismate_mutase_AMBER_PM6.pkl` in the EasyHybrid folder (**File → Open**):

Model	*B. subtilis* CM with chorismate in an active site (at a subunit interface of the trimer) and 1087 water molecules
Force field	AMBER ff99SB
Atoms	8061
Fixed atoms	5673, beyond 14 Å from C5 of the substrate; 2388 free atoms
QC region	chorismate, 24 atoms, **PM6**, charge −2 (two carboxylates)

The QC region is the smallest possible, the substrate alone, which keeps each energy at about 0.03 s: enough for scans, paths and umbrella sampling on a workstation.

The reaction coordinate

RC = d(C5–O7) − d(C1–C9)

Pick	Atom	Index
#1	C5	2034
#2	O7	2037
#3	C9	2039
#4	C1	2026

The pair #1–#2 is the bond that breaks, #3–#4 the bond that forms. In the reactant, C5–O7 = 1.48 Å and C1···C9 = 3.15 Å: RC = −1.68 Å.

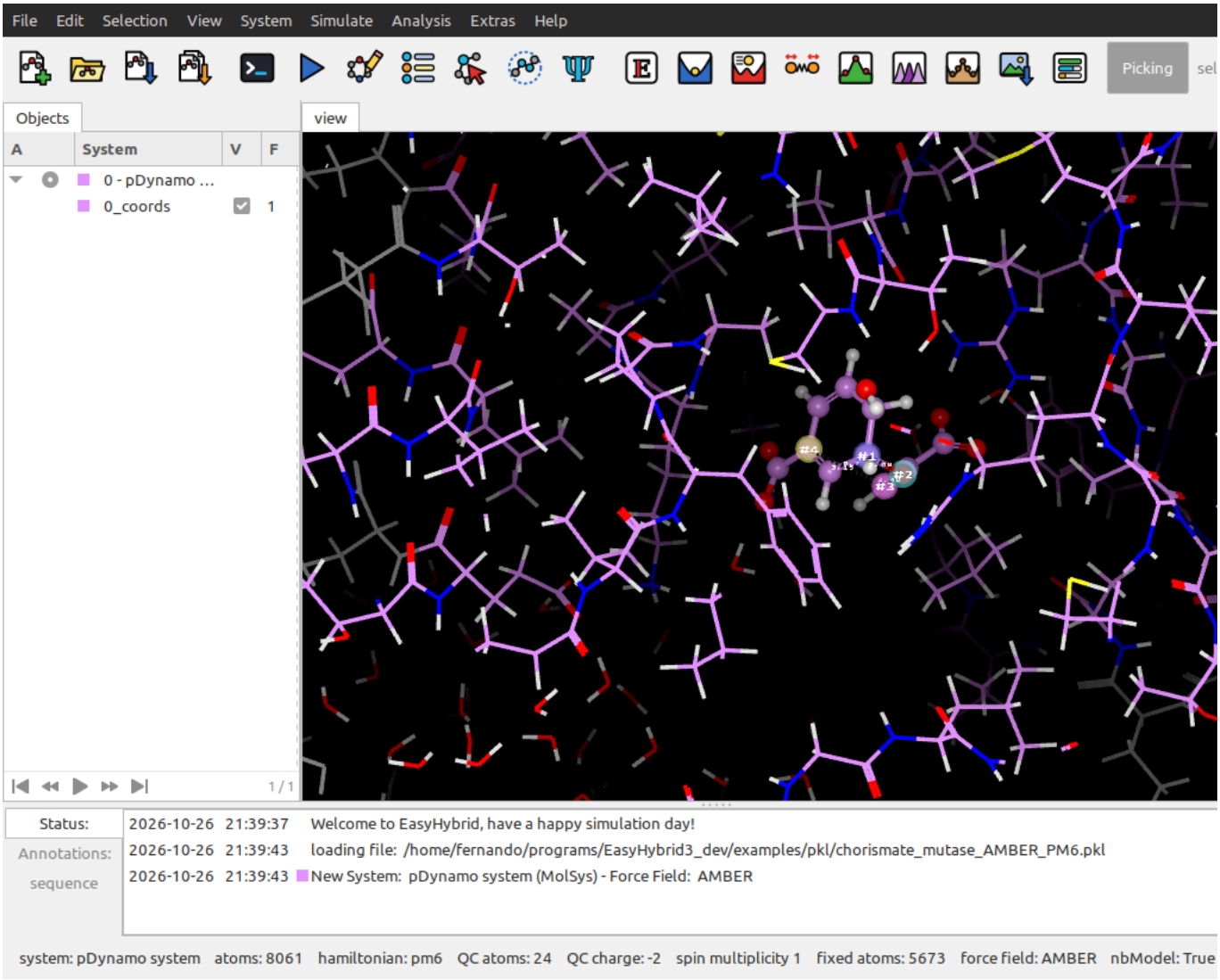


Figure 1. Chorismate (balls and sticks, the QC region) in the active site, with C5, O7, C9 and C1 picked.

The scan

Simulate → Reaction Coordinate Scans, **Coordinate Type** *multiple distance *4 atoms*, and **Import from Picking Selection** with the four atoms picked:

Field	Value
Step Size	0.1 Å
Number of Steps	32 (RC from −1.68 to +1.43 Å)
Force Constant	4000 kJ mol^{-1} Å^{-2}
Coordinates	`0_coords`
Method / Maximum iterations / RMS Grad. Tolerance	Conjugate Gradient / 600 / 0.1
Trajectory Name	`CM_scan`

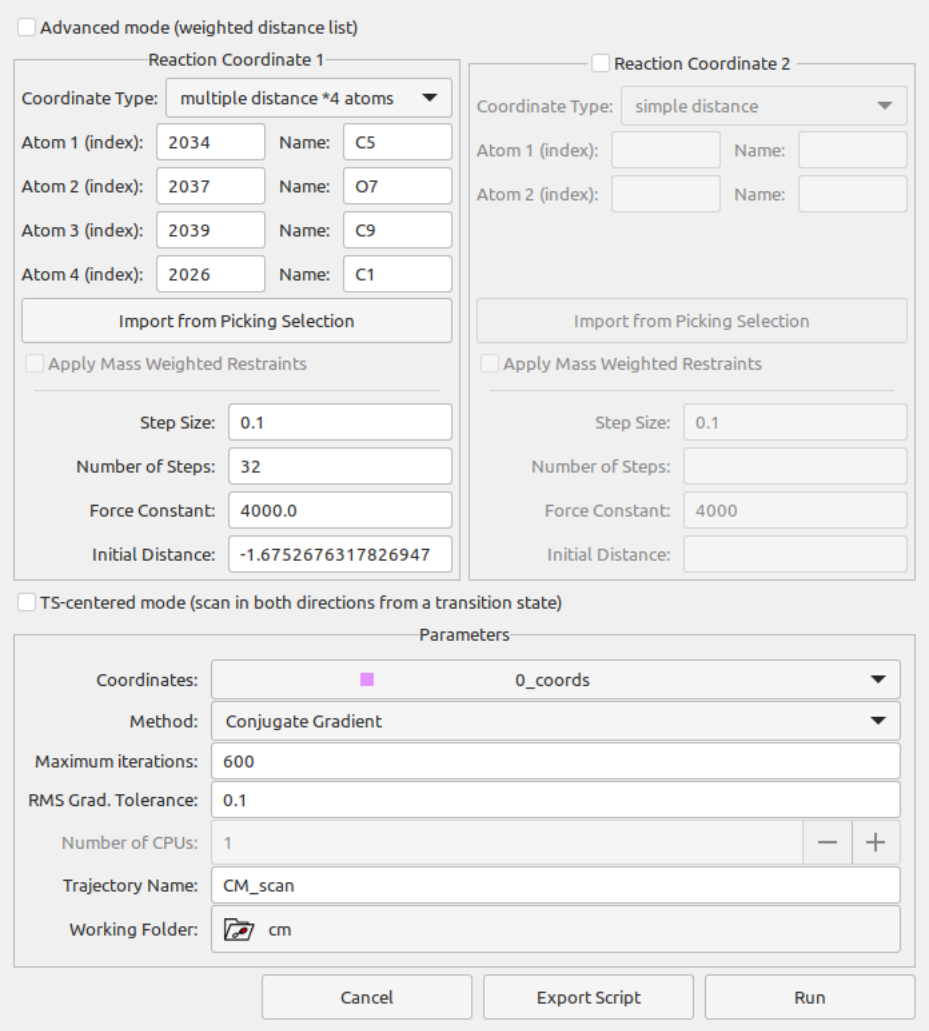


Figure 2. The four-atom coordinate of the Claisen rearrangement.

The scan takes about 1.5 minutes on one core. Import it from the *Finished!* dialog and open **Analysis → PES Analysis**.

Results

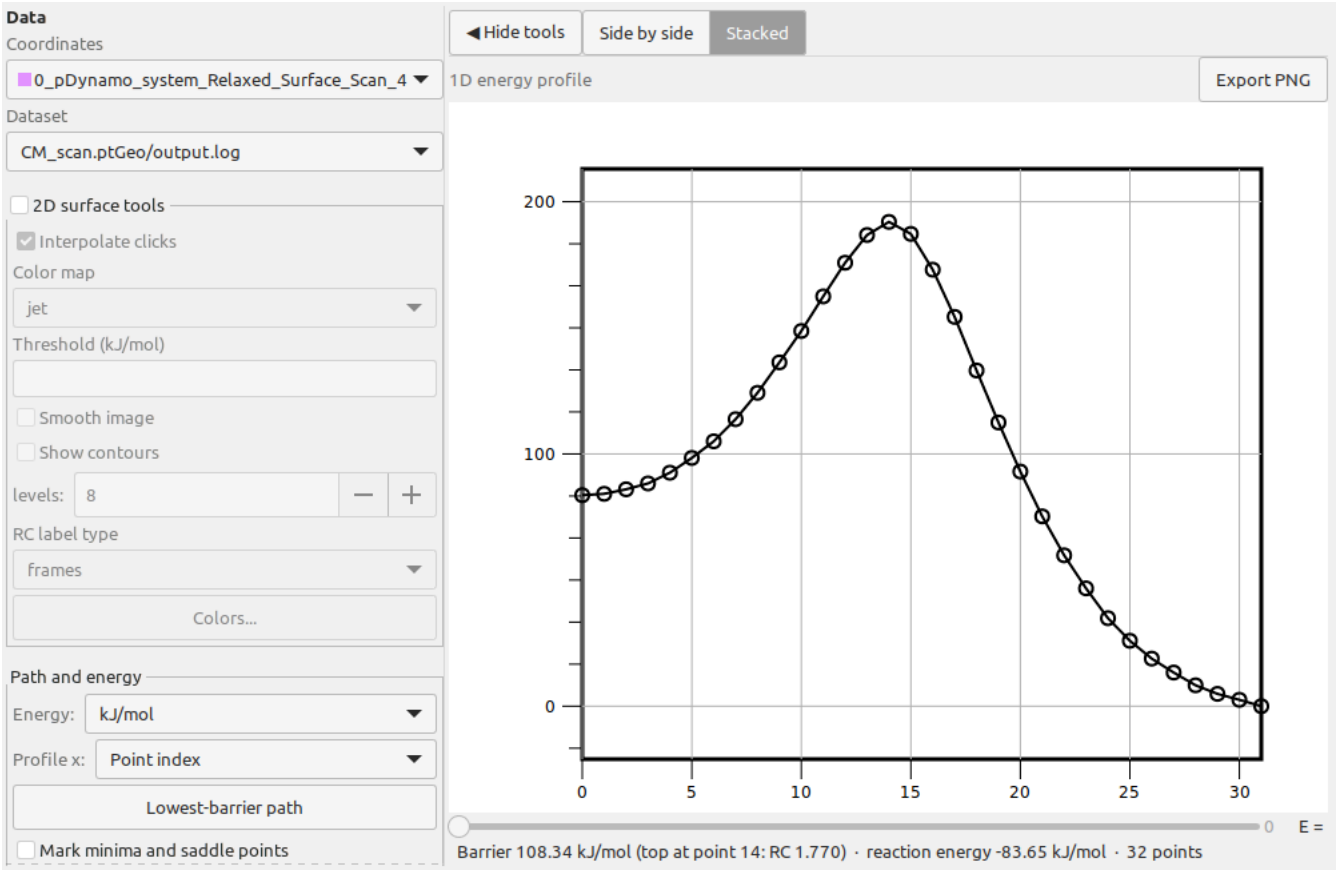


Figure 3. PM6/AMBER energy along the scan (relative to the lowest point, the product).

Point	C5–O7 (Å)	C1–C9 (Å)	E − E(reactant) (kJ/mol)
0	1.48	3.15	0.0 (chorismate)
14	1.77	2.05	**108.4** (highest point)
31	2.99	1.56	−83.6 (prephenate)

At the highest point the C5–O7 bond is stretched by 0.3 Å while the new C1–C9 bond is still 2.05 Å long: an asynchronous transition state, with bond breaking ahead of bond making, as found in most calculations of this reaction.

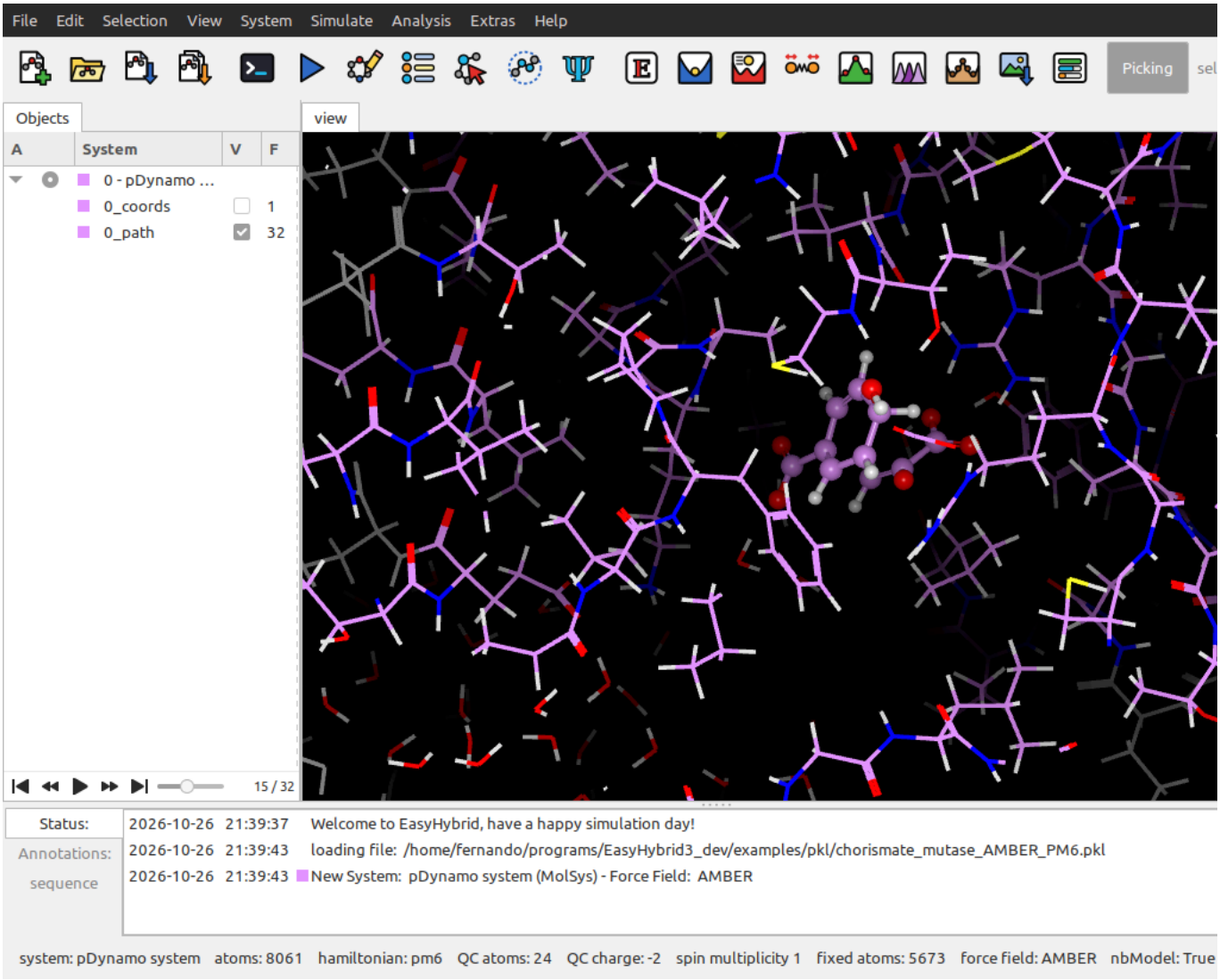


Figure 4. Point 14, the highest point of the scan.

The barrier, 108 kJ/mol (26 kcal/mol), is too high: the experimental free energy of activation in *B. subtilis* CM is about 15.4 kcal/mol (64 kJ/mol). Part of the difference is the method (PM6 overestimates this barrier; higher-level energies at the PM6 structures, with **Analysis → Energy Refinement**, usually lower it), part is the nature of a potential energy scan: one minimized structure per point, with the enzyme around it frozen in one conformation.

Notes

- **Asymmetric coordinate.** The forming and breaking bonds change at different rates: the RC steps of 0.1 Å are not uniform steps of either distance (C5–O7 changes by 0.01 Å per step at the start and by 0.1 Å at the end).
- **Product.** The energy is still decreasing at the last point (C1–C9 = 1.56 Å): optimize the last structure without restraints for the product minimum.
- Next: the NEB path, which needs no coordinate.
- Reference: P. Kast, M. Asif-Ullah, N. Jiang, D. Hilvert, *Proc. Natl. Acad. Sci. USA* **1996**, 93, 5043 (kinetics of *B. subtilis* CM); H. Guo, Q. Cui, W. N. Lipscomb, M. Karplus, *Proc. Natl. Acad. Sci. USA* **2001**, 98, 9032 (QC/MM free energy of the reaction).

EasyHybrid 3.2.0 · built 2026-10-29 from `dev` @ `3539da4` · https://github.com/ferbachega/EasyHybrid3